

RECEIVING REPORT

Receiver No.: 18498509

Document Date: 04/30/2025
Posting Date: 04/30/2025
Expiration Date:

P.O.#: 5300130329
Item #: 2
Batch#: ACE0129356

Insp. Lot: 000001310029
Plant: ACE
Plant Description: Clearfield-Solumina
Vendor Name: HEXCEL CORP
Vendor Addr: 1913 N KING ST
SEGUIN
TX 78155

Quality Inspection

Unloading Point:
Vendor No.: 90069901
Buyer Name: Keith Larsen
SQAR Code:
Project ID: ASBUX

Test Lab:
WBS:

Inspection Lot/Type:

Rev No.: 01

Material: MSRR4040-VAFR

Received UOM: LB
Base UOM: LB
Base UOM Received Qty: 2979.210

Vendor Mtl. Number:
Mtl. Description: SATIN WEAVE CARBON FIBRE REINFORCED BISM
Mtl. Specification:

Alt. Material #:

Matl. Basic Data Text:
MSRR4040-VAFR:SATIN WEAVE CARBON FIBRE REINFORCED BISM

Bill of Lading: 92SPS055644
Receipt Created by: N50973
Head text:

Delivery Note: 92SPS055644
Received Qty: 2979.210
Receipt Delivered To: _____

WAREHOUSE INFORMATION



Plt: ACE



Part Number: MSRR4040-VAFR



Qty: 2979.21



Receiver No: 18498509

Storage Bin:



Order Number:

Warehouse: LIN1



Stock Type: Q



WBS Element: A-46Z01-NW.N46Z01000713

Inventory WBS: Y-ROLLS

Bin:

Storage Location: ACED

Item Text:

SATIN WEAVE CARBON FIBRE REINFORCED BISMALEIMIDE PREIMPREGN
ATTENTION: The revision level for this part is: 4
NG PROJECT: 46001.IN.INV.0RAW, PRIME CONTRACT:
Ship via: Best Way.

Qty. Insp.	Insp. Stamp No.	By Inspector	Qty. Rej.
Rej. No.	Stocked By	Date Stkd.	Stk. Loc.



29 April 2025

CERTIFICATE OF ANALYSIS

Customer: Northrop Grumman Systems Corporation

Purchase Order No: 5300130329 Part Number: MSRR4040-VAFR

Material: F3C 433/M65: 38%; 285AW; 50"

Specification: MSRR4040, Issue 4, dated November 2012.

Certificate of Conformance: Hexcel lot 92M0103014 has been manufactured and tested in conformance with specification MSRR4040, Issue 4, dated November 2012.

Lot No: 92M0103014 Date of Manufacture: 10 April 2025

Storage Life: 12 months from date of manufacture when stored at 0°.

Resin Lot No: 92M0104653 Manufactured by: Hexcel

Glass Lot No: 31W0026929 Manufactured by: Hexcel

- I. Prepreg Physical, Chemical and Mechanical Properties
See Certificate of Analysis Report.
- II. Individual Spool Physical Properties
See In Process Control Acceptance Testing Data.

Note: See Packing List for date of shipment, actual quantity and spools shipped.

Brandi Bond





Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: MSRR4040 Rev. Issue 4
Configuration Number: CN1301
Material Type: F3C 433/M65;38%;285AW;50"
Lot Number: 92M0103014
Order Number: 92M0103014

Report Printed: 23-Apr-2025 2:12:10AM

Status: PASS
Approval: VANZEBKD

1. Tensile, SBS, and TG Panel

Dimensional Fiber Volume

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103014001/001	Laminate Density By Immersion Avg (g/cc)		1.5137
	Laminate Density By Immersion Std (g/cc)		0.0046
	Laminate Density By Immersion CoV		0.31
	Dimensional Fiber Volume (%)		54.84 - 54.19 - 53.80
	Dimensional Fiber Volume Avg (%)	50 - 56	54
	Dimensional Fiber Volume Std (%)		0.525
	Dimensional Fiber Volume CoV		0.97
	Void Volume (%)		0.9 - 1.2 - 1.2
	Void Volume Avg (%)	MAX= 2	1
	Void Volume Std (%)		0.157
92M0103014021/002	Laminate Density By Immersion Avg (g/cc)		1.4882
	Laminate Density By Immersion Std (g/cc)		0.0017
	Laminate Density By Immersion CoV		0.11
	Dimensional Fiber Volume (%)		52.49 - 51.98 - 52.20
	Dimensional Fiber Volume Avg (%)	50 - 56	52
	Dimensional Fiber Volume Std (%)		0.256
	Dimensional Fiber Volume CoV		0.49
	Void Volume (%)		2.3 - 2.3 - 2.4
	Void Volume Avg (%)	MAX= 2	2
	Void Volume Std (%)		0.045
92M0103014035/003	Laminate Density By Immersion Avg (g/cc)		1.5064
	Laminate Density By Immersion Std (g/cc)		0.0012
	Laminate Density By Immersion CoV		0.08
	Dimensional Fiber Volume (%)		52.91 - 52.94 - 53.52
	Dimensional Fiber Volume Avg (%)	50 - 56	53
	Dimensional Fiber Volume Std (%)		0.344
	Dimensional Fiber Volume CoV		0.65
	Void Volume (%)		1.1 - 1.2 - 1.3
	Void Volume Avg (%)	MAX= 2	1
	Void Volume Std (%)		0.089
	Void Volume CoV		7.35

Short Beam Shear Stress-Warp-Ambient



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0103014**
Order Number: **92M0103014**

Report Printed: 23-Apr-2025 2:12:10AM
Status: **PASS**
Approval: **VANZEBKD**

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103014001/001	SBS Strength (MPa)		84.2 - 85.6 - 80.9 - 89.5 - 84.7
	SBS Strength Avg (MPa)	MIN= 65	85
	SBS Strength Std (MPa)		3.09
	SBS Strength CoV		3.63
92M0103014021/002	SBS Strength (MPa)		73.5 - 73.1 - 69.5 - 73.2 - 79.0
	SBS Strength Avg (MPa)	MIN= 65	74
	SBS Strength Std (MPa)		3.39
	SBS Strength CoV		4.61
92M0103014035/003	SBS Strength (MPa)		82.2 - 85.0 - 85.5 - 86.4 - 76.3
	SBS Strength Avg (MPa)	MIN= 65	83
	SBS Strength Std (MPa)		4.09
	SBS Strength CoV		4.92

Short Beam Shear Stress-Warp-150C(302F)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103014001/001	SBS Strength (MPa)		66.7 - 68.0 - 69.5 - 66.0 - 66.6
	SBS Strength Avg (MPa)	MIN= 50	67
	SBS Strength Std (MPa)		1.40
	SBS Strength CoV		2.08
92M0103014021/002	SBS Strength (MPa)		69.3 - 69.8 - 70.2 - 71.2 - 71.2
	SBS Strength Avg (MPa)	MIN= 50	70
	SBS Strength Std (MPa)		0.85
	SBS Strength CoV		1.22
92M0103014035/003	SBS Strength (MPa)		64.8 - 66.7 - 66.7 - 68.8 - 67.3
	SBS Strength Avg (MPa)	MIN= 50	67
	SBS Strength Std (MPa)		1.46
	SBS Strength CoV		2.19

Tg By DMA

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103014001/001	Tg by DMA in °C (° C)		266.3
	Tg by DMA in °C Avg (° C)	MIN= 225.0	266.3
92M0103014021/002	Tg by DMA in °C (° C)		266.9
	Tg by DMA in °C Avg (° C)	MIN= 225.0	266.9
92M0103014035/003	Tg by DMA in °C (° C)		270.8
	Tg by DMA in °C Avg (° C)	MIN= 225.0	270.8



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0103014**
Order Number: **92M0103014**

Report Printed: 23-Apr-2025 2:12:10AM

Status: **PASS**
Approval: **VANZEBKD**

Calculated Thickness/Ply

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103014001/001	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.30
	Calculated Thickness/Ply StDev (mm)		0.0025
	Calculated Thickness/Ply CoV		1.27
92M0103014021/002	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.30
	Calculated Thickness/Ply StDev (mm)		0.0025
	Calculated Thickness/Ply CoV		1.16
92M0103014035/003	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.30
	Calculated Thickness/Ply StDev (mm)		0.0025
	Calculated Thickness/Ply CoV		1.03

Tensile Stress, Modulus, and Elongation-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0103014001/001	Norm Tensile Strength (MPa)		950.4 - 966.2 - 945.6 - 901.0 - 935.3
	Norm Tensile Strength Avg (MPa)	MIN= 700	940
	Norm Tensile Strength Std (MPa)		24.31
	Norm Tensile Strength CoV		2.59
	Norm Modulus (GPa)		53.7 - 60.4 - 65.1 - 65.3 - 68.3
	Norm Modulus Avg (GPa)	MIN= 55	63
	Norm Modulus Std (GPa)		5.702
	Norm Modulus CoV		9.11
92M0103014021/002	Norm Tensile Strength (MPa)		885.0 - 896.2 - 886.2 - 913.3 - 892.3
	Norm Tensile Strength Avg (MPa)	MIN= 700	895
	Norm Tensile Strength Std (MPa)		11.42
	Norm Tensile Strength CoV		1.28
	Norm Modulus (GPa)		60.7 - 61.1 - 47.0 - 77.2 - 51.0
	Norm Modulus Avg (GPa)	MIN= 55	59
	Norm Modulus Std (GPa)		11.652
	Norm Modulus CoV		19.62
92M0103014035/003	Norm Tensile Strength (MPa)		957.9 - 965.1 - 976.2 - 944.2 - 1002.2
	Norm Tensile Strength Avg (MPa)	MIN= 700	969
	Norm Tensile Strength Std (MPa)		21.84
	Norm Tensile Strength CoV		2.25
	Norm Modulus (GPa)		73.1 - 45.5 - 59.6 - 60.0 - 39.3
	Norm Modulus Avg (GPa)	MIN= 55	55
	Norm Modulus Std (GPa)		13.334
	Norm Modulus CoV		24.03



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: MSRR4040 Rev. Issue 4
Configuration Number: CN1301
Material Type: F3C 433/M65;38%;285AW;50"
Lot Number: 92M0103014
Order Number: 92M0103014

Report Printed: 23-Apr-2025 2:12:10AM

Status: PASS
Approval: VANZEBKD

2. PHYSICAL PROPERTIES

Volatile content @100°C in pan

Roll	Test/Result:	Requirements:	Values:
92M0103014001	Volatile content @100°C in a Pan (%)		0.4 - 0.4
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	0
92M0103014035	Volatile content @100°C in a Pan (%)		0.2 - 0.2
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	0

3. PHYSICAL TACK

Tack

Roll	Test/Result:	Requirements:	Values:
92M0103014001	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014003	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014005	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014007	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014009	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014011	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014013	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014016	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014018	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014020	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014021	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014023	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014025	Tack Time to Failure (min.)	MIN= 30	30 - 30



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0103014**
Order Number: **92M0103014**

Report Printed: 23-Apr-2025 2:12:10AM

Status: **PASS**
Approval: **VANZEBKD**

Tack (Continued)

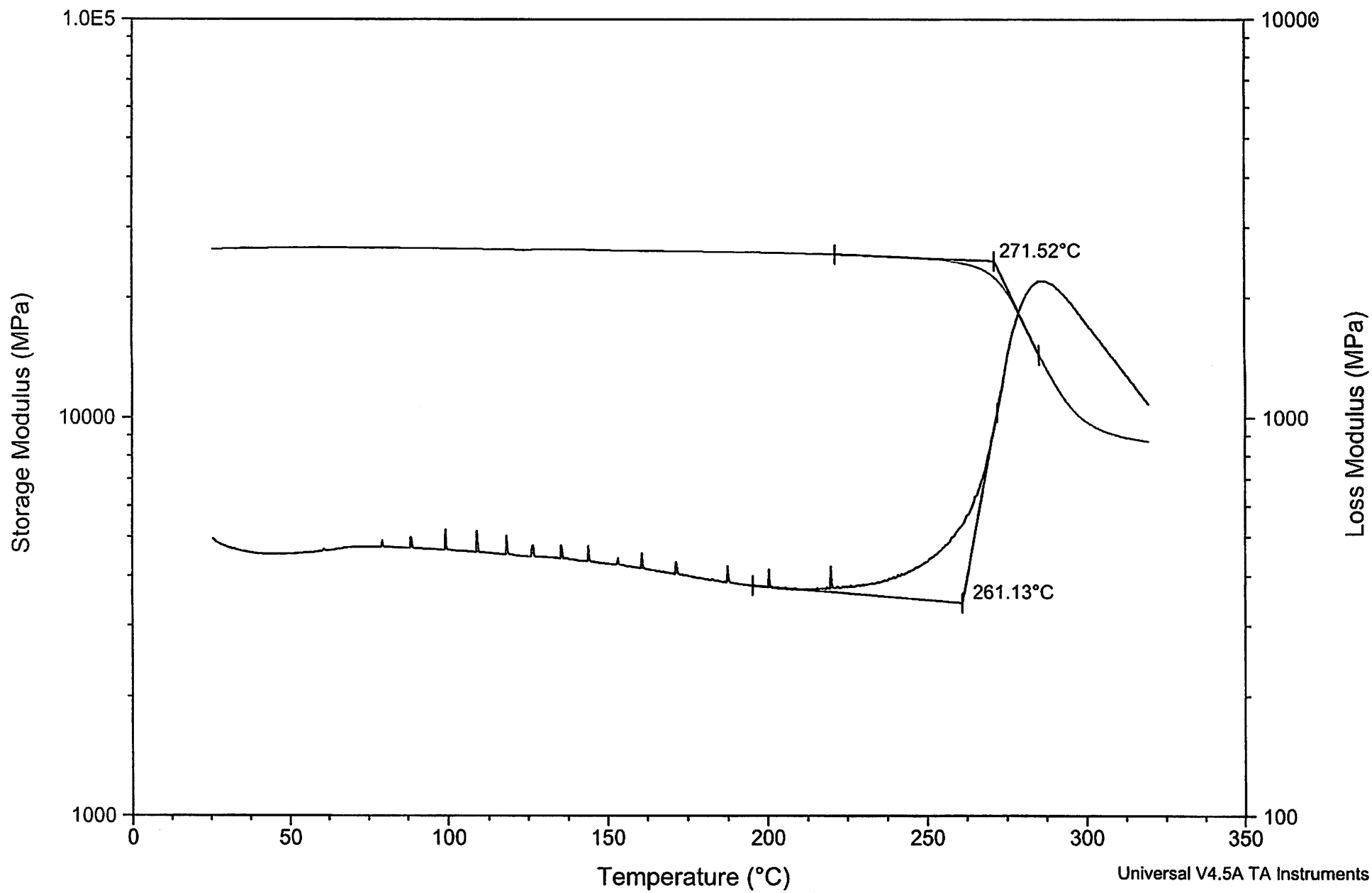
Roll	Test/Result:	Requirements:	Values:
92M0103014025	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014027	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014029	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014031	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0103014033	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30

End of report

Sample: 92M0103014-001 SP#001
Size: 34.9200 x 9.9200 x 2.4300 mm
Method: Temperature Ramp

DMA

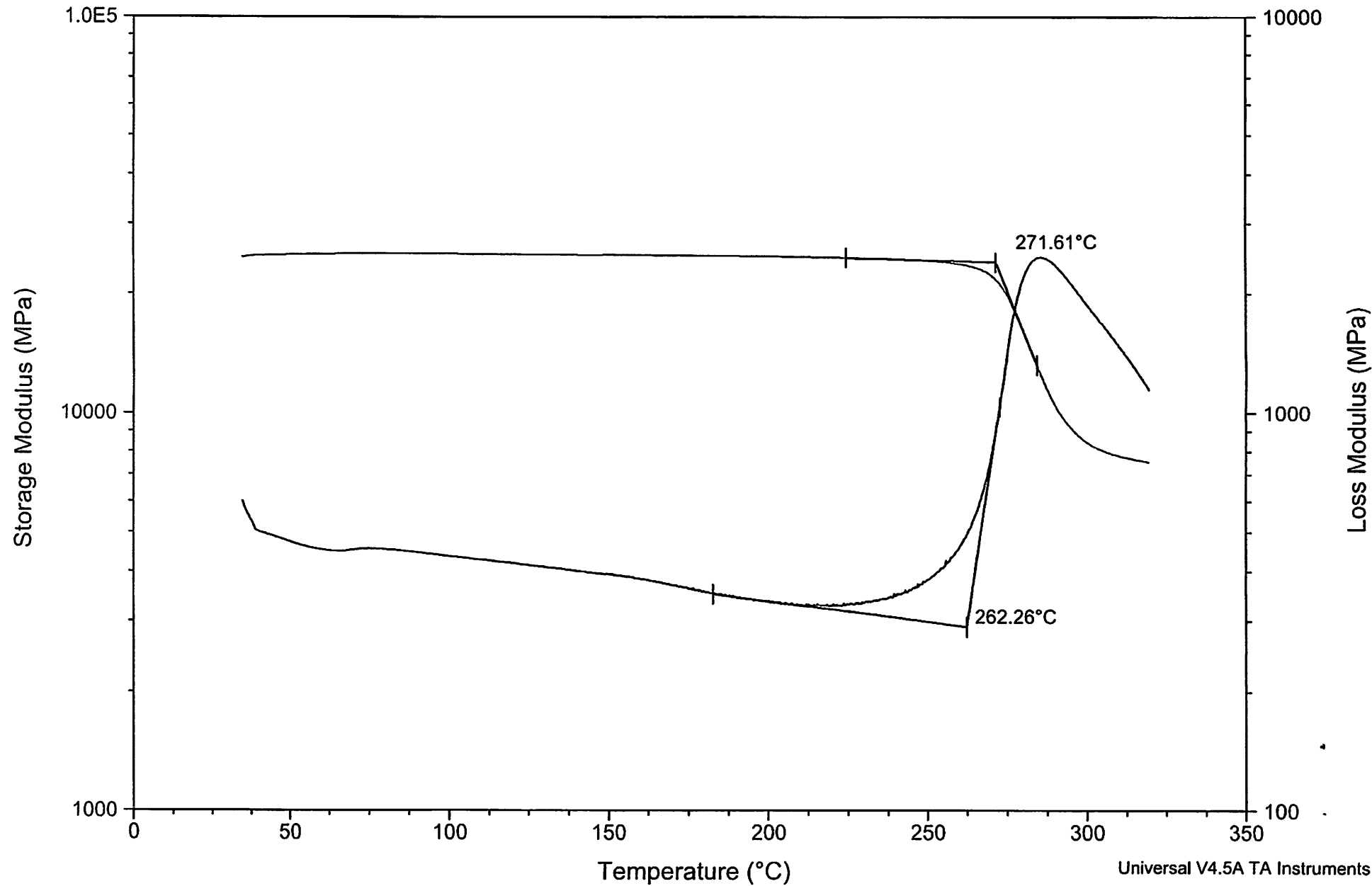
File: \\...\\DMA Q800-2357\\M65\\92M0103014.001
Operator: ak
Run Date: 21-Apr-2025 08:04
Instrument: DMA Q800 V21.3 Build 96



Sample: 92M0103014-002 SP#021
Size: 34.9200 x 9.9200 x 2.5000 mm
Method: Temperature Ramp

DMA

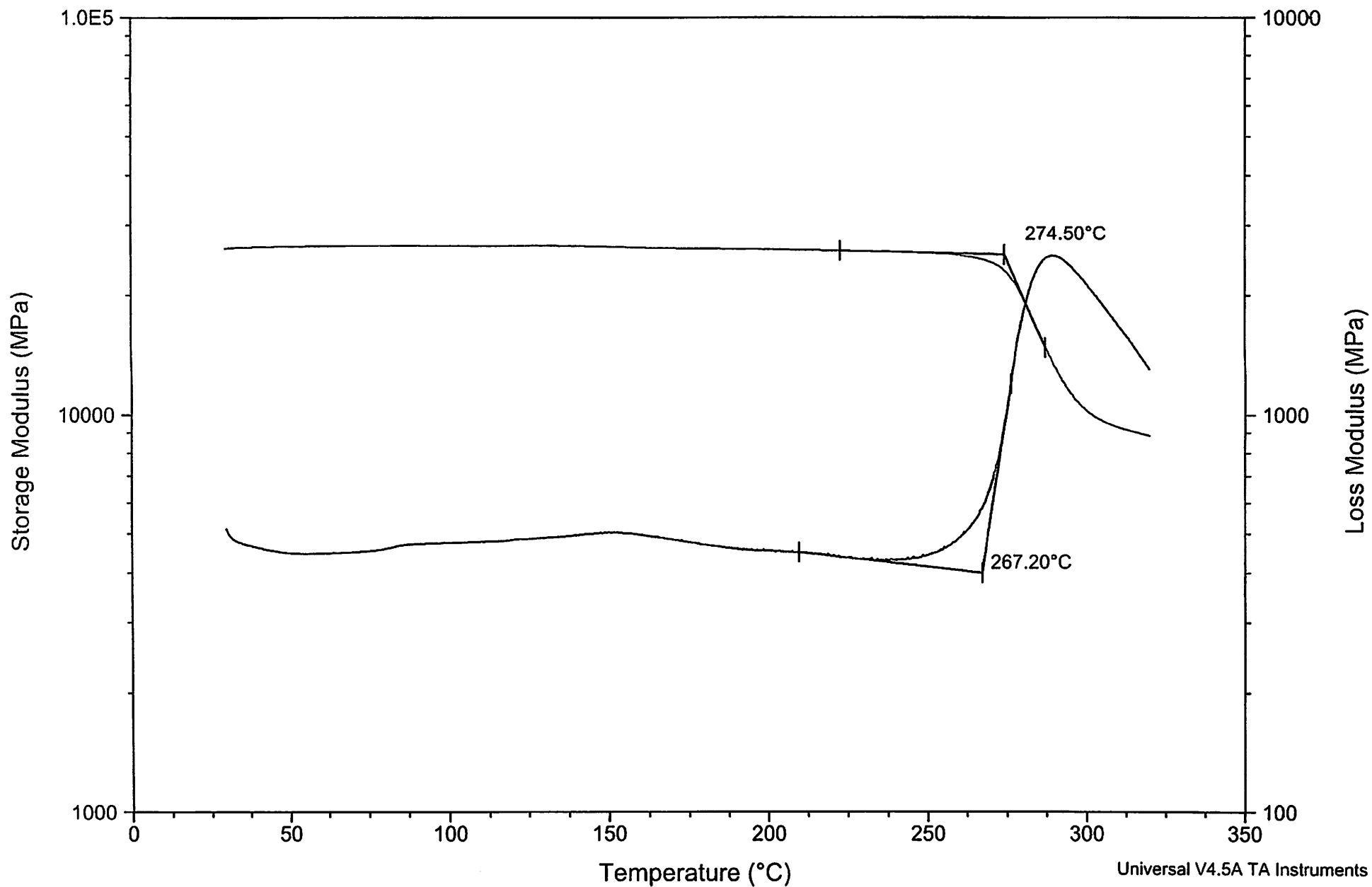
File: \\...\\DMA Q800-2357\\M65\\92M0103014.002
Operator: ak
Run Date: 21-Apr-2025 09:30
Instrument: DMA Q800 V21.3 Build 96



Sample: 92M0103014-003 SP#035
Size: 34.9200 x 9.9300 x 2.4400 mm
Method: Temperature Ramp

DMA

File: \\...\\DMA Q800-2357\\M65\\92M0103014.003
Operator: ak
Run Date: 21-Apr-2025 10:55
Instrument: DMA Q800 V21.3 Build 96



IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

RC / AWT Report for lot: 92M0103014

4/25/2025

Page 1 of 5

Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
92M 0103014001	0	1	4.5869	2.8397	38.1	284	458.7			ELKINSTX	04/10/2025
		2	4.6293	2.8427	38.6	284	462.9			ELKINSTX	04/10/2025
		3	4.6209	2.8446	38.4	284	462.1			ELKINSTX	04/10/2025
		Avg.	4.6124		38	284	461	50.25	A	ELKINSTX	04/10/2025
92M 0103014002	0	1	4.6105	2.8115	39.0	281	461.0			ELKINSTX	04/10/2025
		2	4.6322	2.8275	39.0	283	463.2			ELKINSTX	04/10/2025
		3	4.5817	2.8193	38.5	282	458.2			ELKINSTX	04/10/2025
		Avg.	4.6081		39	282	461	50.25	A	ELKINSTX	04/10/2025
92M 0103014003	0	1	4.6093	2.8227	38.8	282	460.9			ELKINSTX	04/10/2025
		2	4.5952	2.8145	38.8	281	459.5			ELKINSTX	04/10/2025
		3	4.5552	2.8207	38.1	282	455.5			ELKINSTX	04/10/2025
		Avg.	4.5866		39	282	459	50.20	A	ELKINSTX	04/10/2025
92M 0103014004	0	1	4.6451	2.7898	39.9	279	464.5			ELKINSTX	04/10/2025
		2	4.6711	2.7930	40.2	279	467.1			ELKINSTX	04/10/2025
		3	4.6419	2.7946	39.8	280	464.2			ELKINSTX	04/10/2025
		Avg.	4.6527		40	279	465	50.25	A	ELKINSTX	04/10/2025
92M 0103014005	0	1	4.5903	2.8268	38.4	283	459.0			ELKINSTX	04/10/2025
		2	4.6167	2.8353	38.6	284	461.7			ELKINSTX	04/10/2025
		3	4.5847	2.8225	38.4	282	458.5			ELKINSTX	04/10/2025
		Avg.	4.5972		38	283	460	50.20	A	ELKINSTX	04/10/2025
92M 0103014006	0	1	4.6782	2.8067	40.0	281	467.8			ELKINSTX	04/10/2025
		2	4.6689	2.8005	40.0	280	466.9			ELKINSTX	04/10/2025
		3	4.5277	2.7881	38.4	279	452.8			ELKINSTX	04/10/2025
		Avg.	4.6249		40	280	462	50.20	A	ELKINSTX	04/10/2025
92M 0103014007	0	1	4.6432	2.8218	39.2	282	464.3			ELKINSTX	04/10/2025
		2	4.6388	2.8238	39.1	282	463.9			ELKINSTX	04/10/2025
		3	4.6197	2.8190	39.0	282	462.0			ELKINSTX	04/10/2025
		Avg.	4.6339		39	282	463	50.20	A	ELKINSTX	04/10/2025
92M 0103014008	0	1	4.5920	2.8319	38.3	283	459.2			ELKINSTX	04/10/2025
		2	4.6201	2.8334	38.7	283	462.0			ELKINSTX	04/10/2025
		3	4.5455	2.8129	38.1	281	454.6			ELKINSTX	04/10/2025
		Avg.	4.5859		38	283	459	50.25	A	ELKINSTX	04/10/2025
92M 0103014009	0	1	4.5643	2.7924	38.8	279	456.4			ELKINSTX	04/10/2025
		2	4.5966	2.7922	39.3	279	459.7			ELKINSTX	04/10/2025
		3	4.5657	2.7927	38.8	279	456.6			ELKINSTX	04/10/2025
		Avg.	4.5755		39	279	458	50.25	A	ELKINSTX	04/10/2025
92M 0103014010	0	1	4.5454	2.8078	38.2	281	454.5			ELKINSTX	04/10/2025
		2	4.6151	2.8263	38.8	283	461.5			ELKINSTX	04/10/2025
		3	4.5269	2.8369	37.3	284	452.7			ELKINSTX	04/10/2025
		Avg.	4.5625		38	282	456	50.25	A	ELKINSTX	04/10/2025
92M 0103014011	0	1	4.5766	2.7977	38.9	280	457.7			MEKDARJX	04/10/2025
		2	4.6031	2.7926	39.3	279	460.3			MEKDARJX	04/10/2025
		3	4.5434	2.8641	37.0	286	454.3			MEKDARJX	04/10/2025

ACCEPTANCE TESTING DATA

Page 2 of 5

Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
=====											
			Avg. 4.5744		38	282	457	50.25	A	MEKDARJX	04/10/2025
=====											
92M	0	1	4.5762	2.8181	38.4	282	457.6			MEKDARJX	04/10/2025
0103014012		2	4.5956	2.8255	38.5	283	459.6			MEKDARJX	04/10/2025
		3	4.5874	2.8146	38.6	282	458.7			MEKDARJX	04/10/2025
			Avg. 4.5864		38	282	459	50.25	A	MEKDARJX	04/10/2025
=====											
92M	0	1	4.6314	2.8640	38.2	286	463.1			MEKDARJX	04/10/2025
0103014013		2	4.6540	2.8548	38.7	286	465.4			MEKDARJX	04/10/2025
		3	4.6305	2.8699	38.0	287	463.0			MEKDARJX	04/10/2025
			Avg. 4.6386		38	286	464	50.30	A	MEKDARJX	04/10/2025
=====											
92M	0	1	4.5915	2.8213	38.6	282	459.2			MEKDARJX	04/10/2025
0103014014		2	4.5970	2.8083	38.9	281	459.7			MEKDARJX	04/10/2025
		3	4.6251	2.8132	39.2	281	462.5			MEKDARJX	04/10/2025
			Avg. 4.6045		39	281	460	50.30	A	MEKDARJX	04/10/2025
=====											
92M	0	1	4.5238	2.8255	37.5	283	452.4			MEKDARJX	04/10/2025
0103014015		2	4.5318	2.8260	37.6	283	453.2			MEKDARJX	04/10/2025
		3	4.1763	2.8184	32.5	282	417.6			MEKDARJX	04/10/2025
			Avg. 4.4106		36	282	441	50.30	A	MEKDARJX	04/10/2025
=====											
92M	0	1	4.5453	2.8029	38.3	280	454.5			MEKDARJX	04/10/2025
0103014016		2	4.5845	2.8076	38.8	281	458.4			MEKDARJX	04/10/2025
		3	4.5886	2.7826	39.4	278	458.9			MEKDARJX	04/10/2025
			Avg. 4.5728		39	280	457	50.30	A	MEKDARJX	04/10/2025
=====											
92M	0	1	4.5231	2.8526	36.9	285	452.3			MEKDARJX	04/10/2025
0103014017		2	4.5980	2.8566	37.9	286	459.8			MEKDARJX	04/10/2025
		3	4.5889	2.8496	37.9	285	458.9			MEKDARJX	04/10/2025
			Avg. 4.5700		38	285	457	50.30	A	MEKDARJX	04/10/2025
=====											
92M	0	1	4.6093	2.8486	38.2	285	460.9			MEKDARJX	04/10/2025
0103014018		2	4.6069	2.8516	38.1	285	460.7			MEKDARJX	04/10/2025
		3	4.5408	2.8272	37.7	283	454.1			MEKDARJX	04/10/2025
			Avg. 4.5857		38	284	459	50.30	A	MEKDARJX	04/10/2025
=====											
92M	0	1	4.5406	2.8235	37.8	282	454.1			MEKDARJX	04/10/2025
0103014019		2	4.6530	2.8420	38.9	284	465.3			MEKDARJX	04/10/2025
		3	4.5989	2.8306	38.5	283	459.9			MEKDARJX	04/10/2025
			Avg. 4.5975		38	283	460	50.30	A	MEKDARJX	04/10/2025
=====											
92M	0	1	4.5449	2.8295	37.7	283	454.5			MEKDARJX	04/10/2025
0103014020		2	4.6147	2.8520	38.2	285	461.5			MEKDARJX	04/10/2025
		3	4.6263	2.8393	38.6	284	462.6			MEKDARJX	04/10/2025
			Avg. 4.5953		38	284	460	50.25	A	MEKDARJX	04/10/2025
=====											
92M	0	1	4.6462	2.8503	38.7	285	464.6			MEKDARJX	04/10/2025
0103014021		2	4.6610	2.8631	38.6	286	466.1			MEKDARJX	04/10/2025
		3	4.5973	2.8504	38.0	285	459.7			MEKDARJX	04/10/2025
			Avg. 4.6348		38	285	464	50.20	A	MEKDARJX	04/10/2025
=====											
92M	0	1	4.6311	2.8948	37.5	290	463.1			MEKDARJX	04/10/2025
0103014022											

IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

RC / AWT Report for lot: 92M0103014

4/25/2025

Page 3 of 5

Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
92M 0103014022	0	2	4.6426	2.8909	37.7	289	464.3			MEKDARJX	04/10/2025
		3	4.6036	2.8754	37.5	288	460.4			MEKDARJX	04/10/2025
		Avg.	4.6258		38	289	463	50.30	A	MEKDARJX	04/10/2025
92M 0103014023	0	1	4.5936	2.8805	37.3	288	459.4			MEKDARJX	04/10/2025
		2	4.6099	2.9015	37.1	290	461.0			MEKDARJX	04/10/2025
		3	4.6005	2.8772	37.5	288	460.0			MEKDARJX	04/10/2025
		Avg.	4.6013		37	289	460	50.25	A	MEKDARJX	04/10/2025
92M 0103014024	0	1	4.6406	2.8909	37.7	289	464.1			MEKDARJX	04/10/2025
		2	4.6495	2.8992	37.6	290	465.0			MEKDARJX	04/10/2025
		3	4.6295	2.8756	37.9	288	463.0			MEKDARJX	04/10/2025
		Avg.	4.6399		38	289	464	50.25	A	MEKDARJX	04/10/2025
92M 0103014025	0	1	4.6227	2.8785	37.7	288	462.3			MEKDARJX	04/10/2025
		2	4.6433	2.8788	38.0	288	464.3			MEKDARJX	04/10/2025
		3	4.6116	2.8501	38.2	285	461.2			MEKDARJX	04/10/2025
		Avg.	4.6259		38	287	463	50.20	A	MEKDARJX	04/10/2025
92M 0103014026	0	1	4.6279	2.8454	38.5	284	462.8			MEKDARJX	04/10/2025
		2	4.6431	2.8524	38.6	285	464.3			MEKDARJX	04/10/2025
		3	4.6289	2.8420	38.6	284	462.9			MEKDARJX	04/10/2025
		Avg.	4.6333		39	285	463	50.20	A	MEKDARJX	04/10/2025
92M 0103014027	0	1	4.5563	2.8553	37.3	286	455.6			MEKDARJX	04/10/2025
		2	4.5635	2.8697	37.1	287	456.4			MEKDARJX	04/10/2025
		3	4.5673	2.8490	37.6	285	456.7			MEKDARJX	04/10/2025
		Avg.	4.5624		37	286	456	50.20	A	MEKDARJX	04/10/2025
92M 0103014028	0	1	4.5067	2.8636	36.5	286	450.7			MEKDARJX	04/10/2025
		2	4.5116	2.8785	36.2	288	451.2			MEKDARJX	04/10/2025
		3	4.5106	2.8619	36.6	286	451.1			MEKDARJX	04/10/2025
		Avg.	4.5096		36	287	451	50.25	A	MEKDARJX	04/10/2025
92M 0103014029	0	1	4.5844	2.8809	37.2	288	458.4			MEKDARJX	04/10/2025
		2	4.5949	2.8961	37.0	290	459.5			MEKDARJX	04/10/2025
		3	4.5695	2.8702	37.2	287	457.0			MEKDARJX	04/10/2025
		Avg.	4.5829		37	288	458	50.25	A	MEKDARJX	04/10/2025
92M 0103014030	0	1	4.5601	2.8289	38.0	283	456.0			MEKDARJX	04/10/2025
		2	4.5496	2.8104	38.2	281	455.0			MEKDARJX	04/10/2025
		3	4.5553	2.8182	38.1	282	455.5			MEKDARJX	04/10/2025
		Avg.	4.5550		38	282	456	50.30	A	MEKDARJX	04/10/2025
92M 0103014031	0	1	4.5574	2.8203	38.1	282	455.7			MEKDARJX	04/10/2025
		2	4.5061	2.8153	37.5	282	450.6			MEKDARJX	04/10/2025
		3	4.5094	2.8133	37.6	281	450.9			MEKDARJX	04/10/2025
		Avg.	4.5243		38	282	452	50.30	A	MEKDARJX	04/10/2025
92M 0103014032	0	1	4.5648	2.8219	38.2	282	456.5			MEKDARJX	04/10/2025
		2	4.5029	2.8226	37.3	282	450.3			MEKDARJX	04/10/2025
		3	4.5433	2.8200	37.9	282	454.3			MEKDARJX	04/10/2025

ACCEPTANCE TESTING DATA

Page 4 of 5

Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
			Avg. 4.5370		38	282	454	50.30	A	MEKDARJX	04/10/2025
92M 0103014033	0	1	4.6323	2.8847	37.7	288	463.2			MEKDARJX	04/10/2025
		2	4.6653	2.8894	38.1	289	466.5			MEKDARJX	04/10/2025
		3	4.6384	2.8784	37.9	288	463.8			MEKDARJX	04/10/2025
			Avg. 4.6453		38	288	464	50.25	A	MEKDARJX	04/10/2025
92M 0103014034	0	1	4.5931	2.8554	37.8	286	459.3			MEKDARJX	04/10/2025
		2	4.6281	2.8636	38.1	286	462.8			MEKDARJX	04/10/2025
		3	4.6111	2.8633	37.9	286	461.1			MEKDARJX	04/10/2025
			Avg. 4.6108		38	286	461	50.25	A	MEKDARJX	04/10/2025
92M 0103014035	0	1	4.6009	2.8257	38.6	283	460.1			MEKDARJX	04/10/2025
		2	4.6188	2.8332	38.7	283	461.9			MEKDARJX	04/10/2025
		3	4.6144	2.8280	38.7	283	461.4			MEKDARJX	04/10/2025
			Avg. 4.6114		39	283	461	50.20	A	MEKDARJX	04/10/2025

IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

Lot Averages Report for Lot: 92M0103014

04/25/2025

Page 5 of 5

Spool	Retest:	RC	AWT	PAW	Stamp
92M0103014001	0	38	284	461	A
92M0103014002	0	39	282	461	A
92M0103014003	0	39	282	459	A
92M0103014004	0	40	279	465	A
92M0103014005	0	38	283	460	A
92M0103014006	0	40	280	462	A
92M0103014007	0	39	282	463	A
92M0103014008	0	38	283	459	A
92M0103014009	0	39	279	458	A
92M0103014010	0	38	282	456	A
92M0103014011	0	38	282	457	A
92M0103014012	0	38	282	459	A
92M0103014013	0	38	286	464	A
92M0103014014	0	39	281	460	A
92M0103014015	0	36	282	441	A
92M0103014016	0	39	280	457	A
92M0103014017	0	38	285	457	A
92M0103014018	0	38	284	459	A
92M0103014019	0	38	283	460	A
92M0103014020	0	38	284	460	A
92M0103014021	0	38	285	464	A
92M0103014022	0	38	289	463	A
92M0103014023	0	37	289	460	A
92M0103014024	0	38	289	464	A
92M0103014025	0	38	287	463	A
92M0103014026	0	39	285	463	A
92M0103014027	0	37	286	456	A
92M0103014028	0	36	287	451	A
92M0103014029	0	37	288	458	A
92M0103014030	0	38	282	456	A
92M0103014031	0	38	282	452	A
92M0103014032	0	38	282	454	A
92M0103014033	0	38	288	464	A
92M0103014034	0	38	286	461	A
92M0103014035	0	39	283	461	A
Averages:		38.2	283.8	459.0	



Product Certification

Page 1 of 2

COA Number:	SEG-COA-28153	Prod Order:	DOM:	Description:
Version:	1	31W0026929	13-Mar-2025	F3C433(FBL2)-50"
Issue Date:	21-Mar-2025	Weave Pattern:	5 Harness Satin	

Specifications
F3C433(FBL2) - GREIGE - 50Inch - HRMS 111546 Rev. 5 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex
F3C433(FBL2) - GREIGE - 50Inch - MSRR4040 Issue 4 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex
F3C433(FBL2) - GREIGE - 50Inch - MSRR9992 Issue 12 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

Rolls Tested

Fabric Testing Results

Serial#:	Prod Order:	Property:	Specification:	Result:
31G0000356911	31W0026929	Width	49.5 - 50.5 in	50.3
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000356912		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	287
31G0000356913		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.4
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000356914		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000356915		Width	49.5 - 50.5 in	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5

This is to certify that material shipped against this order meets the specification(s) listed above.

This report should only be reproduced in its entirety
1913 N. King Street, Seguin, TX 78155
Telephone (830) 379-1580 Fax (830) 379-3814
www.hexcel.com


Michael Galvan
Certification Technician



Product Certification

Page 2 of 2

31G0000356915	31W0026929	Dried FAW Avg	278 - 292 g/m ²	284
31G0000356916		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000356917		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	283

Certification Exceptions/Additional Information

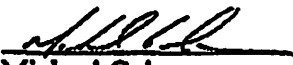
QMS07045, REV. 1, SEC. 5E

WARP: 240920210

FILL: 230920220

This is to certify that material shipped against this order meets the specification(s) listed above.

This report should only be reproduced in its entirety
1913 N. King Street, Seguin, TX 78155
Telephone (830) 379-1580 Fax (830) 379-3814
www.hexcel.com


Michael Galvan
Certification Technician



Product Certification

Page 1 of 3

COA Number:	SEG-COA-28153	Prod Order:	DOM:	Description:
Version:	1	31W0026929	13-Mar-2025	F3C433(FBL2)-50"
Issue Date:	31-Mar-2025	Weave Pattern:	5 Harness Satin	

Specifications

F3C433(FBL2) - GREIGE - 50Inch - HRMS 111546 Rev. 5 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex
F3C433(FBL2) - GREIGE - 50Inch - MSRR4040 Issue 4 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex
F3C433(FBL2) - GREIGE - 50Inch - MSRR9992 Issue 12 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

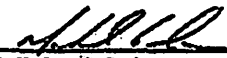
Rolls Tested

Fabric Testing Results

Serial#:	Prod Order:	Property:	Specification:	Result:
31G0000356918	31W0026929	Width	49.5 - 50.5 In	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000356921		Width	49.5 - 50.5 In	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.4
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000356922		Width	49.5 - 50.5 In	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	286
31G0000356923		Width	49.5 - 50.5 In	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000356924		Width	49.5 - 50.5 In	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.5

This is to certify that material shipped against this order meets the specification(s) listed above.

This report should only be reproduced in its entirety
1913 N. King Street, Seguin, TX 78155
Telephone (830) 379-1580 Fax (830) 379-3814
www.hexcel.com


Michael Galvan
Certification Technician



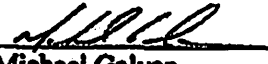
Product Certification

Page 2 of 3

31G0000356924	31W0026929	Dried FAW Avg	278 - 292 g/m ²	286
31G0000356925		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	286
31G0000356926		Width	49.5 - 50.5 in	50.1
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000356927		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000356928		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000356929		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000356930		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	287
31G0000356931		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.4
		Dried FAW Avg	278 - 292 g/m ²	286
31G0000356932		Width	49.5 - 50.5 in	50.3
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5

This is to certify that material shipped against this order meets the specification(s) listed above.

This report should only be reproduced in its entirety
1913 N. King Street, Seguin, TX 78155
Telephone (830) 379-1580 Fax (830) 379-3814
www.hexcel.com


Michael Galvan
Certification Technician



Product Certification

Page 3 of 3

31G0000356932	31W0026929	Dried FAW Avg	278 - 292 g/m ²	283
31G0000356933		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000356934		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	283

Certification Exceptions/Additional Information

QMS07045, REV. 1, SEC. 5E

WARP: 240920210

FILL: 230920220

This is to certify that material shipped against this order meets the specification(s) listed above.

This report should only be reproduced in its entirety
1913 N. King Street, Seguin, TX 78155
Telephone (830) 379-1580 Fax (830) 379-3814
www.hexcel.com


Michael Galvan
Certification Technician

TEIJIN

TEIJIN CARBON AMERICA, INC.

121 Cardiff Valley Road
Rockwood, TN 37854
Tel. (865) 354-8408
Fax (865) 354-8409

Certificate of Analysis

QAP 004.P2 Rev M

Customer: Hexcel Reinforcement Corp (TX) Contact: Patrick Speckermann;
Amy Shehane; Jerry Tschospe; Greg Early Email/Fax: Patrick.Speckermann@hexcel.com;
amy.shehane@hexcel.com;
jerry.tschospe@; Greg.Early@

Customer PO#: 31PO0013019-1 Order: 58424 Ship Date: 3/15/2024

Customer P/N: C1805820N1 Product: Tenax™-E HTA40 E13 3K 200tex

Part Number: G305303 Quantity: 1178.3 lb Certifies To: MS-X-CTC-105 Rev B;
MSRR9317/11 Issue 1

Lot Number: 230920220

We are pleased to advise that the product (s) listed above has been released for shipment. We certify that this lot of Carbon Fiber has been tested by standard test methods and determined to have the following properties:

	Average	cv%
Tensile Strength, MPA (ksi):	4288.0 (619)	3.3
Tensile Modulus, GPa (msi):	237.5 (34.4)	1.0
Elongation, %:	1.70	3.3
Linear Density (without size), g/m:	0.2013	0.7
Size, %:	1.28	6.9
Fiber Density, g/cc:	1.77	
Filament Diameter, µm:	7	
Twist, turn/m:	Never Twisted	
Date of Manufacture:	September 2023	
Manufacturing Location:	Oberbruch, Heinsberg, Germany	

Quality Assurance

3/13/2024

These commodities, technology or software are controlled by U. S. Export Administration Regulation. Diversion contrary to U. S. Law is prohibited.

If goods are not received in a reasonable time or if there are any questions, please call 1-800-282-3001 (or (865) 354-8408),
Teijin Carbon America, Inc. 121 Cardiff Valley Road, Rockwood; TN 37854.

TEIJIN

TEIJIN CARBON AMERICA, INC.

121 Cardiff Valley Road
Rockwood, TN 37854
Tel. (865) 354-8408
Fax (865) 354-8409

Certificate of Analysis

QAP 004.F2 Rev M

Customer: Hexcel Reinforcement Corp (TX) Contact: Patrick Spieckermann;
Amy Shehane; Jerry Tschoepe; Greg Early Email/Fax: Patrick.Spieckermann@hexcel.com;
amy.shehane@hexcel.com;
jerry.tschoepe@; Greg.Early@

CustomerPO#: 31PC0013988 RELEASE 1 Order: SH001381 Ship Date: 3/8/2025

Customer P/N: n/a Product: Tenax™-E HTA40 E13 3K 200tex

Part Number: G305303 Quantity: 3000.000 kg Certifies To: MS-X-GTC-105 Rev B; MSRR9317/11
Issue 1

Lot Number: 240920210

We are pleased to advise that the product (s) listed above has been released for shipment. We certify that this lot of Carbon Fiber has been tested by standard test methods and determined to have the following properties:

	<u>Average</u>	<u>CV%</u>
Tensile Strength, MPA (ksi):	4213.2 (611)	3.8
Tensile Modulus, GPA (msi):	238.1 (34.5)	0.9
Elongation, %:	1.67	3.7
Linear Density (without size), g/m:	0.2003	0.8
Size, %:	1.28	7.5
Fiber Density, g/cc:	1.77	
Filament Diameter, µm:	7	
Twist, turn/m:	Never Twisted	
Date of Manufacture:	September 2024	
Manufacturing Location:	Oberbruch, Heinsberg, Germany	


Quality Assurance 3/3/2025

These commodities, technology or software are controlled by U. S. Export Administration Regulation. Diversion contrary to U. S. Law is prohibited.

If goods are not received in a reasonable time or if there are any questions, please call 1-800-282-3001 (or (865) 354-8408).
Teijin Carbon America, Inc. 121 Cardiff Valley Road, Rockwood, TN 37854.

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014001	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
66	50	0	0.0	FUZZBALL
78	50	0	0.0	FUZZBALL
99	50	0	0.0	FUZZBALL
108	50	0	0.0	FUZZBALL
125	50	0	0.0	FUZZBALL
131	50	0	0.0	FUZZBALL
144	50	2	0.7	FLOAT
179	50	2	0.7	FLOAT
184	50	0	0.0	FUZZBALL
189	50	0	0.0	FUZZBALL
205	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
225	224	A	0.148%	4/10/2025

Inspection Stamp	Date
	4/10/2025

ROLL CONDITION LOG

Batch Number 92M0103014	Serial Number 92M0103014002	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
12	50	0	0.0	FUZZBALL
91	50	1	0.3	SLUB
121	50	0	0.0	FUZZBALL

Final Footage 210	Net Feet 209	Credit Method A	Defect % 0.040%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
---	--------------------------

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014003	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
27	50	0	0.0	FUZZBALL
41	50	2	0.7	FUZZBALL
78	50	0	0.0	FUZZBALL
90	50	0	0.0	FUZZBALL
155	50	0	0.0	FUZZBALL
162	50	0	0.0	FUZZBALL
222	50	2	0.7	CREASE

Final Footage	Net Feet	Credit Method	Defect %	Date
230	229	A	0.145%	4/10/2025

Inspection Stamp	Date
	4/10/2025

ROLL CONDITION LOG

Batch Number 92M0103014	Serial Number 92M0103014004	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
67	50	0	0.0	FUZZBALL
99	50	0	0.0	FUZZBALL
128	50	12	4.2	SNAG
130	50	0	0.0	FUZZBALL
135	50	12	4.2	SNAG
145	50	0	0.0	FUZZBALL
164	50	0	0.0	FUZZBALL
170	50	0	0.0	FUZZBALL
186	50	0	0.0	CREASE

Final Footage 218	Net Feet 216	Credit Method A	Defect % 0.917%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
---	--------------------------

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014005	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
102	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
225	225	A	0.000%	4/10/2025

Inspection Stamp	Date
	4/10/2025

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014006	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
37	50	0	0.0	FUZZBALL
49	50	2	0.7	FUZZBALL
88	50	2	0.7	FUZZBALL
100	50	0	0.0	FUZZBALL
128	50	0	0.0	FUZZBALL

APR 14 2025



Final Footage	Net Feet	Credit Method	Defect %	Date
225	224 222	A	0.148%	4/10/2025

Inspection Stamp	Date
	4/10/2025

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014007	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
6	50	2	0.7	FUZZBALL
42	50	0	0.0	FUZZBALL
47	50	0	0.0	FUZZBALL
112	50	0	0.0	FUZZBALL
129	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
225	224	A	0.074%	4/10/2025

Inspection Stamp	Date
	4/10/2025

Batch Number 92M0103014	Serial Number 92M0103014008	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
5	50	1	0.3	RESIN LEAN
25	50	1	0.3	MISPICK
115	50	0	0.0	FUZZBALL

Final Footage 224	Net Feet 223	Credit Method A	Defect % 0.074%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
--	--------------------------

Batch Number 92M0103014	Serial Number 92M0103014009	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
53	50	12	4.2	RESIN RICH
126	50	0	0.0	FUZZBALL
133	50	48	16.7	CRIMP
172	50	1	0.3	RESIN LEAN

Final Footage 228	Net Feet 222	Credit Method A	Defect % 2.230%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
---	--------------------------

Batch Number 92M0103014	Serial Number 92M0103014010	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
5	50	0	0.0	FUZZBALL
51	50	0	0.0	FUZZBALL
75	50	0	0.0	FUZZBALL
102	50	0	0.0	FUZZBALL
162	50	2	0.7	MISPICK

Final Footage 220	Net Feet 219	Credit Method A	Defect % 0.076%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
---	--------------------------

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014011	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
25	50	0	0.0	FUZZBALL
42	50	0	0.0	FUZZBALL
75	50	0	0.0	FUZZBALL
93	50	6	2.1	FLOAT
150	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
225	224	A	0.222%	4/10/2025

Inspection Stamp	Date
	4/10/2025

Batch Number 92M0103014	Serial Number 92M0103014012	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
114	50	0	0.0	FUZZBALL
158	50	0	0.0	FUZZBALL
164	50	0	0.0	FUZZBALL

Final Footage 217	Net Feet 217	Credit Method A	Defect % 0.000%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
---	--------------------------

Batch Number 92M0103014		Serial Number 92M0103014013		Fiber F3C 433	Resin M65	Prepreg Width 50
Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks		
185	50	0	0.0	FUZZBALL		

Final Footage 228	Net Feet 228	Credit Method A	Defect % 0.000%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
--	--------------------------

Batch Number 92M0103014	Serial Number 92M0103014014	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
----------------------------	--------------------------------------	---------------------------------------	--	----------------------------

Final Footage 221	Net Feet 221	Credit Method A	Defect % 0.000%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
--	--------------------------

Batch Number 92M0103014	Serial Number 92M0103014015	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
15	50	72	25.0	RESIN LEAN
29	50	0	0.0	FUZZBALL

Final Footage 110	Net Feet 104	Credit Method A	Defect % 5.455%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
---	--------------------------

Batch Number 92M0103014	Serial Number 92M0103014016	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
9	50	0	0.0	0
28	50	0	0.0	FUZZBALL
38	50	0	0.0	FUZZBALL
43	50	12	4.2	RESIN LEAN
157	50	0	0.0	FUZZBALL

Final Footage 223	Net Feet 222	Credit Method A	Defect % 0.448%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
--	--------------------------

Batch Number 92M0103014		Serial Number 92M0103014017		Fiber F3C 433	Resin M65	Prepreg Width 50
Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks		
42	50	0	0.0	FUZZBALL		

Final Footage 225	Net Feet 225	Credit Method A	Defect % 0.000%	Date 4/10/2025
----------------------	-----------------	--------------------	--------------------	-------------------

Inspection Stamp	Date
	4/10/2025

Batch Number 92M0103014		Serial Number 92M0103014018		Fiber F3C 433	Resin M65	Prepreg Width 50
Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks		
3	50	1	0.3	RESIN LEAN		
166	50	0	0.0	FUZZBALL		

Final Footage 222	Net Feet 221	Credit Method A	Defect % 0.038%	Date 4/10/2025
----------------------	-----------------	--------------------	--------------------	-------------------

Inspection Stamp	Date
	4/10/2025

Batch Number 92M0103014	Serial Number 92M0103014019	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
--------------------	-----------------------------	------------------------------	-------------------------------	---------------------

Final Footage 225	Net Feet 225	Credit Method A	Defect % 0.000%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
---	--------------------------

Batch Number 92M0103014		Serial Number 92M0103014020		Fiber F3C 433	Resin M65	Prepreg Width 50
Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks		
29	50	0	0.0	FUZZBALL		

Final Footage 161	Net Feet 161	Credit Method A	Defect % 0.000%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
---	--------------------------

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014021	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
73	50	0	0.0	FUZZBALL
100	50	0	0.0	FUZZBALL
139	50	0	0.0	FUZZBALL
157	50	0	0.0	FUZZBALL
200	50	24	8.3	CURED RESIN
205	50	1	0.3	RESIN LEAN

Final Footage	Net Feet	Credit Method	Defect %	Date
225	222	A	0.926%	4/10/2025

Inspection Stamp	Date
	4/10/2025

ROLL CONDITION LOG

Batch Number 92M0103014	Serial Number 92M0103014022	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
--------------------	-----------------------------	------------------------------	-------------------------------	---------------------

Final Footage 215	Net Feet 215	Credit Method A	Defect % 0.000%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
--	--------------------------

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014023	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
72	50	0	0.0	FUZZBALL
81	50	0	0.0	FUZZBALL
135	50	1	0.3	RESIN LEAN
140	50	0	0.0	FUZZBALL
150	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
215	214	A	0.039%	4/10/2025

Inspection Stamp	Date
	4/10/2025

ROLL CONDITION LOG

Batch Number 92M0103014	Serial Number 92M0103014024	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
20	50	0	0.0	FUZZBALL
27	50	0	0.0	FUZZBALL
46	50	0	0.0	FUZZBALL
62	50	0	0.0	FUZZBALL
71	50	0	0.0	FUZZBALL
125	50	1	0.3	RESIN LEAN
130	50	0	0.0	FUZZBALL
140	50	0	0.0	FUZZBALL

Final Footage 205	Net Feet 204	Credit Method A	Defect % 0.041%	Date 4/10/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/10/2025
---	--------------------------

ROLL CONDITION LOG

Batch Number 92M0103014	Serial Number 92M0103014025	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
126	50	0	0.0	FUZZBALL

Final Footage 225	Net Feet 225	Credit Method A	Defect % 0.000%	Date 4/11/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/11/2025
--	--------------------------

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014026	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
55	50	0	0.0	FUZZBALL
89	50	0	0.0	FUZZBALL
159	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
225	225	A	0.000%	4/11/2025

Inspection Stamp	Date
	4/11/2025

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014027	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
--------------------	-----------------------------	------------------------------	-------------------------------	---------------------

Final Footage	Net Feet	Credit Method	Defect %	Date
225	225	A	0.000%	4/11/2025

Inspection Stamp	Date
	4/11/2025

ROLL CONDITION LOG

Batch Number 92M0103014	Serial Number 92M0103014028	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
47	50	0	0.0	FUZZBALL
61	50	0	0.0	FUZZBALL
83	50	0	0.0	FUZZBALL

Final Footage 215	Net Feet 215	Credit Method A	Defect % 0.000%	Date 4/11/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp  	Date 4/11/2025
---	--------------------------

ROLL CONDITION LOG

Batch Number 92M0103014	Serial Number 92M0103014029	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
206	50	60	20.8	SPLICE
216	50	12	4.2	SPLICE

Final Footage 225	Net Feet 219	Credit Method A	Defect % 2.667%	Date 4/14/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/14/2025
---	--------------------------

ROLL CONDITION LOG

Batch Number 92M0103014	Serial Number 92M0103014030	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
9	50	0	0.0	FUZZBALL
83	50	0	0.0	FUZZBALL
108	50	0	0.0	FUZZBALL
200	50	0	0.0	FUZZBALL

Final Footage 206	Net Feet 206	Credit Method A	Defect % 0.000%	Date 4/11/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/11/2025
--	--------------------------

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014031	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
28	50	0	0.0	FUZZBALL
55	50	0	0.0	FUZZBALL
74	50	0	0.0	FUZZBALL
85	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
225	225	A	0.000%	4/11/2025

Inspection Stamp	Date
	4/11/2025

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014032	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
11	50	0	0.0	FUZZBALL
53	50	1	0.3	SLUB
76	50	0	0.0	FUZZBALL
150	50	0	0.0	FUZZBALL
184	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
225	224	A	0.037%	4/11/2025

Inspection Stamp	Date
	4/11/2025

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014033	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
14	50	0	0.0	FUZZBALL
45	50	0	0.0	FUZZBALL
95	50	0	0.0	FUZZBALL
105	50	0	0.0	FUZZBALL
182	50	0	0.0	FUZZBALL
219	50	0	0.0	FUZZBALL
225	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
300	300	A	0.000%	4/11/2025

Inspection Stamp	Date
	4/11/2025

ROLL CONDITION LOG

Batch Number 92M0103014	Serial Number 92M0103014034	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
11	50	2	0.7	FUZZBALL
127	50	0	0.0	FUZZBALL
239	50	0	0.0	FUZZBALL

Final Footage 304	Net Feet 303	Credit Method A	Defect % 0.055%	Date 4/11/2025
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 4/11/2025
--	--------------------------

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0103014	92M0103014035	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
--------------------	-----------------------------	------------------------------	-------------------------------	---------------------

Final Footage	Net Feet	Credit Method	Defect %	Date
117	117	A	0.000%	4/11/2025

Inspection Stamp	Date
	4/11/2025